Getting to know your data

There are four main projects that progress sequentially throughout the book. In each subsequent chapter, the code will expand upon the code in previous chapters. We chose these projects to highlight a variety of Data Intelligence Platform features across different ML projects. Specifically, we include streaming data into your lakehouse architecture, forecasting sales, building a **deep learning** (**DL**) model for computer vision, and building a chatbot using **Retrieval Augmented Generation** (**RAG**) techniques. Read through the descriptions of each project to get an idea of what it will cover. If some of the concepts and features are unfamiliar, don't worry! We'll explain them in the following chapters.

Project: Streaming Transactions

This first project is a data solution for the streaming transactions dataset we will generate. The transaction data will include information such as the customer ID and transaction time, which we'll use to simulate transactions streaming in real time; see the sample data in *Figure 1.6*.

Amount	CustomerID	Label	Product	TransactionTimestamp
10.0	1240	0	Product A	2023-12-12T23:36:16.284Z
250.01	1253	0	Product A	2023-12-12T23:36:16.284Z
10.0	1254	0	Product A	2023-12-12T23:36:16.284Z
250.01	1253	0	Product A	2023-12-12T23:36:16.284Z
69.99	1244	0	Product A	2023-12-12T23:36:16.284Z
10.0	1247	0	Product A	2023-12-12T23:36:16.284Z
250.01	1242	0	Product A	2023-12-12T23:36:16.284Z

Figure 1.6 – A sample of the synthetic streaming transactions data

Our goal with this project is to demonstrate how flexible the Data Intelligence Platform is compared to proprietary data warehouses of the past, which were more rigid for data ingestion. Additionally, we want to highlight important Databricks capabilities such as **Spark Structured Streaming**, **Auto Loader**, schema evolution, Delta Live Tables, and **Lakehouse Monitoring**.